

EE114 – Homework 5

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Problem 2.7.1

Setup (Quiz 2.6): Monitor three phone calls; each is voice or data. N = number of voice calls with PMF

$$P_N(n) = \begin{cases} 0.1 & n = 0, \\ 0.3 & n = 1, 2, 3, \\ 0 & \text{otherwise.} \end{cases}$$

Voice calls cost 25 cents each; data calls cost 40 cents each. The total cost is T cents. From Quiz 2.6 part (1): $T = 25N + 40(3 - N) = 120 - 15N$.

(a) Find $E[T]$ using Theorem 2.10 (the derived-variable theorem $E[g(X)] = \sum_x g(x)P_X(x)$):

$$\begin{aligned} E[T] &= E[120 - 15N] = \sum_{n=0}^3 (120 - 15n) P_N(n) \\ &= (120)(0.1) + (105)(0.3) + (90)(0.3) + (75)(0.3) \\ &= 12 + 31.5 + 27 + 22.5 \\ &= \boxed{93 \text{ cents}}. \end{aligned}$$

(b) Find $E[T]$ using Definition 2.14 (standard expected value $E[T] = \sum_t t P_T(t)$):

First we need $P_T(t)$. Since $T = 120 - 15N$:

n	0	1	2	3
$t = 120 - 15n$	120	105	90	75
$P_N(n)$	0.1	0.3	0.3	0.3

So $P_T(t)$ is:

$$P_T(t) = \begin{cases} 0.1 & t = 120, \\ 0.3 & t = 75, 90, 105, \\ 0 & \text{otherwise.} \end{cases}$$

Then:

$$\begin{aligned} E[T] &= \sum_t t P_T(t) = 75(0.3) + 90(0.3) + 105(0.3) + 120(0.1) \\ &= 22.5 + 27 + 31.5 + 12 = \boxed{93 \text{ cents}}. \end{aligned}$$

Both methods give $E[T] = 93$ cents, as expected.

Problem 2.8.3

Setup (Problem 2.4.2): The random variable X has CDF

$$F_X(x) = \begin{cases} 0 & x < -1, \\ 0.2 & -1 \leq x < 0, \\ 0.7 & 0 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

From the CDF we extract the PMF:

$$P_X(x) = \begin{cases} 0.2 & x = -1, \\ 0.5 & x = 0, \\ 0.3 & x = 1, \\ 0 & \text{otherwise.} \end{cases}$$

First compute $E[X]$:

$$E[X] = (-1)(0.2) + (0)(0.5) + (1)(0.3) = -0.2 + 0 + 0.3 = 0.1.$$

Next compute $E[X^2]$:

$$E[X^2] = (-1)^2(0.2) + (0)^2(0.5) + (1)^2(0.3) = 0.2 + 0 + 0.3 = 0.5.$$

Therefore:

$$\text{Var}[X] = E[X^2] - (E[X])^2 = 0.5 - (0.1)^2 = 0.5 - 0.01 = \boxed{0.49}.$$

Problem 2.8.5

X has the binomial PMF $P_X(x) = \binom{4}{x}(1/2)^4$ for $x = 0, 1, 2, 3, 4$.

This is a Binomial($n = 4, p = 1/2$) random variable.

(a) Standard deviation:

For a Binomial(n, p): $E[X] = np = 4(1/2) = 2$, $\text{Var}[X] = np(1 - p) = 4(1/2)(1/2) = 1$.

$$\sigma_X = \sqrt{\text{Var}[X]} = \sqrt{1} = \boxed{1}.$$

(b) We need $P[\mu_X - \sigma_X \leq X \leq \mu_X + \sigma_X] = P[2 - 1 \leq X \leq 2 + 1] = P[1 \leq X \leq 3]$.

$$\begin{aligned} P[1 \leq X \leq 3] &= P_X(1) + P_X(2) + P_X(3) \\ &= \binom{4}{1}\left(\frac{1}{2}\right)^4 + \binom{4}{2}\left(\frac{1}{2}\right)^4 + \binom{4}{3}\left(\frac{1}{2}\right)^4 \\ &= \frac{4 + 6 + 4}{16} = \frac{14}{16} = \boxed{\frac{7}{8} = 0.875}. \end{aligned}$$

Problem 2.8.8

Given X with mean μ_X and variance σ_X^2 , let

$$Y = \frac{1}{\sigma_X}(X - \mu_X).$$

Mean of Y :

$$E[Y] = E\left[\frac{X - \mu_X}{\sigma_X}\right] = \frac{1}{\sigma_X}(E[X] - \mu_X) = \frac{1}{\sigma_X}(\mu_X - \mu_X) = \boxed{0}.$$

Variance of Y :

$$\text{Var}[Y] = \text{Var}\left[\frac{X - \mu_X}{\sigma_X}\right] = \frac{1}{\sigma_X^2} \text{Var}[X - \mu_X] = \frac{1}{\sigma_X^2} \text{Var}[X] = \frac{\sigma_X^2}{\sigma_X^2} = \boxed{1}.$$

(We used $\text{Var}[aX + b] = a^2 \text{Var}[X]$, and shifting by a constant does not change the variance.)

Problem 3.4.2

Y is an exponential random variable with $\text{Var}[Y] = 25$.

For an exponential(λ) RV: $E[Y] = 1/\lambda$ and $\text{Var}[Y] = 1/\lambda^2$.

From $\text{Var}[Y] = 1/\lambda^2 = 25$ we get $\lambda = 1/5$.

(a) PDF of Y :

$$f_Y(y) = \begin{cases} \frac{1}{5} e^{-y/5} & y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

$$\boxed{f_Y(y) = \frac{1}{5} e^{-y/5}, \quad y \geq 0.}$$

(b) $E[Y^2]$:

$$E[Y^2] = \text{Var}[Y] + (E[Y])^2 = 25 + \left(\frac{1}{\lambda}\right)^2 = 25 + 25 = \boxed{50}.$$

(c) $P[Y > 5]$:

$$P[Y > 5] = \int_5^\infty \frac{1}{5} e^{-y/5} dy = e^{-5/5} = e^{-1} = \boxed{e^{-1} \approx 0.3679}.$$

Problem 3.4.4

Y is an Erlang($n = 2, \lambda = 2$) random variable.

The Erlang(n, λ) PDF is

$$f_Y(y) = \frac{\lambda^n y^{n-1} e^{-\lambda y}}{(n-1)!}, \quad y \geq 0.$$

With $n = 2, \lambda = 2$:

$$f_Y(y) = 4y e^{-2y}, \quad y \geq 0.$$

(a) $E[Y] = n/\lambda = 2/2 = \boxed{1}$.

(b) $\text{Var}[Y] = n/\lambda^2 = 2/4 = \boxed{1/2}$.

(c) $P[0.5 \leq Y < 1.5]$:

The CDF of an Erlang(2, λ) is

$$F_Y(y) = 1 - e^{-\lambda y} - \lambda y e^{-\lambda y} = 1 - (1 + \lambda y)e^{-\lambda y}, \quad y \geq 0.$$

With $\lambda = 2$:

$$F_Y(y) = 1 - (1 + 2y)e^{-2y}.$$

Therefore:

$$\begin{aligned} P[0.5 \leq Y < 1.5] &= F_Y(1.5) - F_Y(0.5) \\ &= [1 - (1 + 3)e^{-3}] - [1 - (1 + 1)e^{-1}] \\ &= [1 - 4e^{-3}] - [1 - 2e^{-1}] \\ &= 2e^{-1} - 4e^{-3} \\ &\approx 2(0.36788) - 4(0.04979) \\ &\approx 0.73576 - 0.19915 \\ &= \boxed{0.5366}. \end{aligned}$$

Problem 3.4.7

The PDF of X is

$$f_X(x) = \begin{cases} \frac{1}{2} e^{-x/2} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

This is an exponential distribution with $\lambda = 1/2$.

(a) $P[1 \leq X \leq 2]$:

$$\begin{aligned} P[1 \leq X \leq 2] &= \int_1^2 \frac{1}{2} e^{-x/2} dx = [-e^{-x/2}]_1^2 = -e^{-1} - (-e^{-1/2}) \\ &= e^{-1/2} - e^{-1} \approx 0.6065 - 0.3679 = \boxed{0.2387}. \end{aligned}$$

(b) **CDF:**

$$F_X(x) = \begin{cases} 0 & x < 0, \\ 1 - e^{-x/2} & x \geq 0. \end{cases}$$

$$\boxed{F_X(x) = 1 - e^{-x/2}, \quad x \geq 0.}$$

(c) $E[X] = 1/\lambda = 1/(1/2) = \boxed{2}$.

(d) $\text{Var}[X] = 1/\lambda^2 = 1/(1/4) = \boxed{4}$.

Problem 3.5.1

The peak temperature T is a Gaussian(85, 10) random variable, meaning $\mu = 85$ and $\sigma^2 = 10$ (so $\sigma = \sqrt{10}$).

We standardize: $Z = (T - \mu)/\sigma = (T - 85)/\sqrt{10}$ is $N(0, 1)$.

$P[T > 100]$:

$$\begin{aligned} P[T > 100] &= P\left[Z > \frac{100 - 85}{\sqrt{10}}\right] = P\left[Z > \frac{15}{\sqrt{10}}\right] = P[Z > 4.743] \\ &= Q(4.743) \approx \boxed{1.07 \times 10^{-6}}. \end{aligned}$$

$P[T < 60]$:

$$\begin{aligned} P[T < 60] &= P\left[Z < \frac{60 - 85}{\sqrt{10}}\right] = P\left[Z < \frac{-25}{\sqrt{10}}\right] = P[Z < -7.906] \\ &= Q(7.906) \approx \boxed{1.30 \times 10^{-15}}. \end{aligned}$$

This is essentially zero.

$P[70 \leq T \leq 100]$:

$$\begin{aligned} P[70 \leq T \leq 100] &= P\left[\frac{70 - 85}{\sqrt{10}} \leq Z \leq \frac{100 - 85}{\sqrt{10}}\right] \\ &= P[-4.743 \leq Z \leq 4.743] \\ &= \Phi(4.743) - \Phi(-4.743) \\ &= 1 - 2Q(4.743) \approx 1 - 2(1.07 \times 10^{-6}) \approx \boxed{0.999998}. \end{aligned}$$

Problem 3.5.4

Show that $Q(z) = \frac{1}{2} \operatorname{erfc}\left(\frac{z}{\sqrt{2}}\right)$.

Proof. By definition:

$$Q(z) = \frac{1}{\sqrt{2\pi}} \int_z^\infty e^{-t^2/2} dt.$$

Substitute $u = t/\sqrt{2}$, so $t = \sqrt{2}u$, $dt = \sqrt{2}du$. When $t = z$, $u = z/\sqrt{2}$; when $t \rightarrow \infty$, $u \rightarrow \infty$.

$$\begin{aligned} Q(z) &= \frac{1}{\sqrt{2\pi}} \int_{z/\sqrt{2}}^\infty e^{-(\sqrt{2}u)^2/2} \sqrt{2} du \\ &= \frac{1}{\sqrt{2\pi}} \int_{z/\sqrt{2}}^\infty e^{-u^2} \sqrt{2} du \\ &= \frac{\sqrt{2}}{\sqrt{2\pi}} \int_{z/\sqrt{2}}^\infty e^{-u^2} du \\ &= \frac{1}{\sqrt{\pi}} \int_{z/\sqrt{2}}^\infty e^{-u^2} du. \end{aligned}$$

Now recall $\operatorname{erfc}(w) = \frac{2}{\sqrt{\pi}} \int_w^\infty e^{-x^2} dx$, so

$$\frac{1}{\sqrt{\pi}} \int_{z/\sqrt{2}}^\infty e^{-u^2} du = \frac{1}{2} \cdot \frac{2}{\sqrt{\pi}} \int_{z/\sqrt{2}}^\infty e^{-u^2} du = \frac{1}{2} \operatorname{erfc}\left(\frac{z}{\sqrt{2}}\right).$$

Therefore:

$$\boxed{Q(z) = \frac{1}{2} \operatorname{erfc}\left(\frac{z}{\sqrt{2}}\right)}. \quad \square$$

Problem 3.5.8

We verify that for $x \geq 0$: $\Phi(x) = \frac{1}{2} + \frac{1}{2} \operatorname{erf}\left(\frac{x}{\sqrt{2}}\right)$.

(a) Let $Y \sim N(0, 1/2)$, so $f_Y(y) = \frac{1}{\sqrt{\pi}} e^{-y^2}$.

Compute $F_Y(y)$:

$$F_Y(y) = \int_{-\infty}^y \frac{1}{\sqrt{\pi}} e^{-u^2} du = \frac{1}{\sqrt{\pi}} \int_{-\infty}^y e^{-u^2} du.$$

Split the integral:

$$F_Y(y) = \frac{1}{\sqrt{\pi}} \left(\int_{-\infty}^0 e^{-u^2} du + \int_0^y e^{-u^2} du \right).$$

Since $\int_{-\infty}^0 e^{-u^2} du = \frac{\sqrt{\pi}}{2}$ and $\operatorname{erf}(y) = \frac{2}{\sqrt{\pi}} \int_0^y e^{-u^2} du$:

$$\begin{aligned} F_Y(y) &= \frac{1}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} + \frac{1}{\sqrt{\pi}} \int_0^y e^{-u^2} du \\ &= \frac{1}{2} + \frac{1}{2} \cdot \frac{2}{\sqrt{\pi}} \int_0^y e^{-u^2} du \\ &= \boxed{\frac{1}{2} + \frac{1}{2} \operatorname{erf}(y)}. \end{aligned}$$

(b) If $Y \sim N(0, 1/2)$, then $Z = \sqrt{2}Y \sim N(0, 1)$, i.e. $Y = Z/\sqrt{2}$.

For $z \geq 0$:

$$\begin{aligned} \Phi(z) &= F_Z(z) = P[Z \leq z] = P\left[Y \leq \frac{z}{\sqrt{2}}\right] = F_Y\left(\frac{z}{\sqrt{2}}\right) \\ &= \frac{1}{2} + \frac{1}{2} \operatorname{erf}\left(\frac{z}{\sqrt{2}}\right). \end{aligned}$$

Therefore:

$$\boxed{\Phi(x) = \frac{1}{2} + \frac{1}{2} \operatorname{erf}\left(\frac{x}{\sqrt{2}}\right), \quad x \geq 0.} \quad \square$$